

Pablo Robin Guerrero

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RESEARCH INTERESTS

Representation Learning, Computational Modeling, Communication and Behavior.

EDUCATION

École Normale Supérieure de Paris (ENS-PSL)

Master in Cognitive Science

Paris, France

September 2026 – Present

École Polytechnique Fédérale de Lausanne (EPFL)

Bachelor of Science in Communication Systems

Lausanne, Switzerland

September 2022 – February 2026

Focus on Theoretical Computer Science and Machine Learning

Relevant coursework:

Machine Learning, Natural Language Processing, Algorithms, Data-intensive Systems, Computer Language Processing, Software Construction, Computer Systems, Computer Security and Privacy, Stochastic Models, Probability and Statistics.

Bachelor Project:

Efficient Deployment of Vision-Language Models on Mobile Devices: A Case Study on OnePlus 13R

EXPERIENCES

École Normale Supérieure de Paris (ENS-PSL)

Research Engineer, Neural Circuit dynamics & Behaviour Laboratory

Paris, France

February 2026 - Present

Research topic:

Social and Latent Acoustic Structure in Bottlenose Dolphin Communication: Linking Whistles to Behaviour

Supervisors: Prof. Germán Sumbre and Faadil Mustun

- Investigate **codebook structure** and learn acoustic representations to identify phoneme-like units and characterize the long-term organization of **bottlenose dolphin whistles**.
- Link **acoustic representations to temporal structure** and behavioural context to probe the organization of dolphin vocal sequences.

Tsinghua University (THU)

Research Intern, NICS Lab, Efficient Deep Learning Algorithm Team

Beijing, China

July 2025 - September 2025

Research topic: Model-Level Optimization for Efficient Generative Inference via Latent-Space Routing

Supervisors: Prof. Yu Wang and Dr. Xuefei Ning

- Worked on **latent-space continual learning** for token-level LLM/SLM routing, enabling model adaptation without full retraining and limiting catastrophic forgetting during inference.
- Explored **DPO-based routing models** operating on latent representations.
- Conducted experimental analysis of **efficiency–performance trade-offs** of routing under constrained inference settings.
- Experimental contributions to **Road2Rome**, accepted at **NeurIPS 2025 (Main Conference)**.

École Polytechnique Fédérale de Lausanne (EPFL)

Research Intern, Robust Scalable Systems Software Laboratory

Lausanne, Switzerland

January 2025 - July 2025

Research topic: Efficient Deployment of Vision-Language Models on Mobile Devices

Supervisors: Prof. Sanidhya Kashyap and Yueyang Pan

- Benchmarked deployment frameworks (llama.cpp, MLC-Imp, mllm) on the **OnePlus 13R** with VLMs (LLaVA-1.5 7B, MobileVLM-3B, Imp-v1.5 3B), measuring **latency, throughput, and device thermals**.
- Profiled inference workloads across **CPU, GPU, and NPU**, demonstrating **CPU saturation** and **accelerator underutilization** as the key barriers to real-time on-device generation.

- Results published in the preprint Efficient Deployment of Vision-Language Models on Mobile Devices: A Case Study on OnePlus 13R (arXiv:2507.08505), providing practical **profiling tools** and **deployment guidelines** for future mobile AI systems.

Korea Advanced Institute of Science & Technology (KAIST)

Research Intern, Concurrency & Parallelism Laboratory

Daejeon, South Korea

June 2024 - August 2024

Research topic: PIM-based AI Accelerator and Software Stack for Efficient LLM Inference

Supervisors: Prof. Jeehoon Kang and Haechan An

- Investigated NeuPIMs architecture, achieving up to **3.0x throughput improvement** over GPU systems and **1.6x improvement** over naive NPU+PIM systems on **GPT-3 (7B)**.
- Evaluated AttAcc design, demonstrating **2.81x higher throughput** and **62.6% improved power efficiency** for **LLaMA 65B** compared to GPU-only systems.
- Reproduced experiments on PIM-based systems, uncovering limitations in multi-device scalability and proposing solutions projected to enhance performance by **25%** in multi-device environments.

PUBLICATIONS

Robin Guerrero, P., Pan, Y., and Kashyap, S. Efficient deployment of vision-language models on mobile devices: A case study on oneplus 13r. *arXiv preprint arXiv:2507.08505*, 2025.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, Scala, Kotlin

Frameworks/Libraries: PyTorch, TensorFlow, Hugging Face Transformers, SGLang

Tools: Git, Docker, CI/CD, Linux, LaTeX

Spoken Languages: French (Native), Spanish (Native), English (Full professional proficiency), Korean (Limited working proficiency)

AWARDS AND HONORS

Tsinghua University Visiting Researcher Scholarship (2025)